

Rotation-Equivariant Variational Autoencoders for Histopathology Representation Learning and Downstream Classification

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Abstract—Histopathology slides contain diagnostic tissue patterns without a canonical orientation, motivating rotation-equivariant representation learning. We compared frozen representations from a conventional variational autoencoder (VAE) and a continuous-SO(2)-equivariant VAE trained with the same data and objective. A common multiple-instance learning protocol assessed five-class whole-slide image (WSI) diagnosis; reconstruction and patch-level tissue classification provided complementary evaluations. The equivariant representation achieved higher WSI macro-F1 (0.483 versus 0.389), but the paired 95% interval for the difference included zero. Reconstruction differences were inconclusive, and tissue classification varied with the label budget.

Index Terms—histopathology, variational autoencoder, rotation equivariance, SO(2), whole-slide image classification, representation learning

I. INTRODUCTION

Digital histopathology converts glass slides into whole-slide images whose dimensions commonly reach gigapixel scale. These images contain many local tissue regions, but their diagnostic labels and detailed annotations require specialist review. Representation learning from unlabeled patches offers a way to use this image volume before fitting a smaller supervised model [1]–[3].

Histopathological images also differ from many natural-image datasets because local tissue structures do not have a fixed upright orientation. A tissue pattern can appear at any in-plane angle without changing its biological meaning. Conventional convolutional networks account for this variation mainly through rotated examples or by learning separate filters for different orientations. Rotation-equivariant networks instead constrain intermediate representations to transform according to the same rotation applied to the input [4]–[6]. Rotated appearances of the same morphology are thereby required to follow a prescribed relation in feature space. This relation motivates evaluating the frozen representations under limited label budgets. This construction is relevant to histopathology, where rotation-equivariant classifiers have previously been evaluated for patch-level prediction [7].

An equivariant architecture does not, however, ensure that a trained model will reconstruct images more accurately or provide a better representation for every supervised task. The comparative outcome must be evaluated under a common data

and training protocol. In this study, two denoising VAEs [8], [9] are compared: a conventional residual convolutional model and a model whose convolutional kernels are constrained to be equivariant to the continuous planar rotation group SO(2). Both models produce a spatial Gaussian latent representation and are frozen before any diagnostic or tissue label is used.

The experiment addresses two questions:

- 1) Does the implemented rotation-equivariant VAE yield lower reconstruction error on unseen histopathology patches than the conventional VAE?
- 2) Does its frozen representation yield higher classification macro-F1 under a shared supervised protocol: (a) for five-class WSI diagnosis from bags of latent maps, and (b) for three-class tissue recognition from individual latent maps across label budgets?

We experimentally characterize the conventional and continuous-SO(2) VAEs, then test their frozen representations in two classification settings. WSI diagnosis evaluates the representation after bag-level aggregation. Tissue recognition removes this aggregation stage and tests whether a compact classifier can recover local tissue information from the complete spatial latent map as the amount of labeled training data changes. All comparisons preserve the image populations, spatial latent dimensions, optimization budget and supervised assignments, and uncertainty is evaluated at WSI level.

II. RELATED WORK

A. Representation Learning in Histopathology

Whole-slide images cannot generally be processed at native resolution in one forward pass. A common approach divides each WSI into patches, learns a local representation, and aggregates patch features for slide-level prediction. Tellez *et al.* used neural image compression to learn patch codes before gigapixel prediction [3]. Attention-based multiple instance learning (MIL) provides a related supervised formulation in which a WSI is treated as a bag of instances and only the bag label is required [10]. CLAM combines attention pooling with instance-level clustering [11]. DSMIL combines contrastive patch representation learning with dual-stream aggregation [12], whereas TransMIL models relations among patch instances during aggregation [13]. In the present experiment,

MIL is an evaluation instrument: the comparison changes the frozen patch representation while holding the WSI classifier protocol fixed.

B. Rotation Equivariance

Group-equivariant convolutional networks construct layers that satisfy this property for a selected transformation group [4]. Steerable convolutions extend the construction to continuous rotations by restricting the kernel space according to the representations carried by the input and output fields [5], [6]. When the target function follows the same symmetry, this restriction can reduce the effective hypothesis space [14]. The architecture also changes parameter count and computational work; both are reported alongside model performance. Previous histopathology work evaluated rotation equivariance in supervised patch classifiers [7]. Here the constraint is introduced during unsupervised VAE training and evaluated through frozen representations.

III. MATERIALS AND METHODS

A. Experimental Design

We evaluated the same frozen VAE pair in three tasks. First, image reconstruction measured whether each VAE could recover a clean, unseen patch from its compressed representation. This evaluation uses the encoder and *decoder*; it is separate from supervised classification. Second, a MIL classifier used frozen patch embeddings to predict one diagnosis for each WSI. Third, a patch classifier used the embeddings to predict local tissue type. For the third task, we repeated training with five increasing amounts of labeled patches per class (the *label budgets*). The subsets were nested: each smaller training subset was contained in every larger one. This design compares how the two representations support classification as supervision increases while keeping the validation and test sets fixed.

The VAEs were trained first on unlabeled patches, then frozen at the final checkpoint. Reconstruction passed a patch through each frozen encoder and decoder; classification used only the deterministic posterior-mean embedding. No downstream label updated either VAE. Both VAEs received the same ordered patches, normalization, corruption policy, loss, update count and validation schedule. Downstream pairs shared WSI assignments, patch coordinates, classifier initialization, epoch order and model-selection rule. Thus, the comparison measures the effect of replacing the ordinary VAE with the implemented continuous-rotation architecture under these controls. The two architectures also have different parameter counts and computational costs (Table II), which matter when interpreting why outcomes differ.

B. Data Source and Image Units

The UBC-OCEAN ovarian-cancer training resource contains 538 slide images: 513 conventional whole-slide images (WSIs) and 25 tissue microarray (TMA) images [15]. A WSI shows a large tissue section; a TMA shows multiple small tissue cores in one image. The resource describes WSIs as

large, approximately $20\times$ scans and TMAs as smaller, approximately 4000×4000 -pixel, $40\times$ images. This scale and layout difference motivated using only the 513 non-TMA WSIs in the present experiment. Figure 3 shows a representative image of each type and their original pixel dimensions. For an image of width W and height H , a non-overlapping 256-pixel grid contains approximately $\lfloor W/256 \rfloor \lfloor H/256 \rfloor$ full patches before foreground filtering. At the dimensions quoted by the resource, a 4000×4000 image yields about 225 full patches, whereas a 100000×50000 image yields about 76,000; the latter is an illustrative large-WSI scale, not a count for an individual study slide. The resource also provides 152 supplemental color masks paired with a subset of those WSIs. These masks mark selected tumor, stroma and necrosis regions; they do not annotate every tissue pixel. The other 361 WSIs have no supplemental mask. The diagnostic label of each image is one of clear cell carcinoma (CC), endometrioid carcinoma (EC), high-grade serous carcinoma (HGSC), low-grade serous carcinoma (LGSC), or mucinous carcinoma (MC).

Each WSI was divided into non-overlapping RGB 256×256 -pixel patches. A full WSI has one diagnosis label, so thousands of patches extracted from that slide share the same slide-level label. Local tissue labels, in contrast, come only from eligible regions of the supplemental masks. Splits were made by WSI: no patches from one slide crossed train, validation or test boundaries.

Figure 1 shows a WSI thumbnail and two actual 256×256 patches from the same VAE-validation slide. The thumbnail makes the slide-to-patch scale change visible; model inference used crops from the full-resolution WSI.

The thumbnail, partial mask and overlay in Figure 2 come from a second, training-cohort WSI. They show how supplemental annotations occupy selected regions rather than the full image.

C. Patch Selection and Tissue Annotations

Foreground selection for reconstruction and WSI diagnosis was derived from the image rather than from human masks. Each WSI thumbnail was converted to HSV, and Otsu's threshold was applied to the saturation channel [17]. A full-resolution grid cell was retained when more than 60% of its projected thumbnail area belonged to foreground. The resulting coordinates were used for VAE development, reconstruction and WSI diagnosis without consulting the supplemental tissue masks.

Supplemental masks were used only to construct the tissue-recognition population. Red, green and blue mask regions were interpreted as tumor, stroma and necrosis. The masks are not exhaustive; an unpainted region cannot be treated as a negative class. A patch was therefore eligible for tissue classification only when at least 10% of its area was annotated and at least 95% of that annotated area belonged to one tissue class. The resulting task classifies high-purity annotated regions and does not constitute dense pixel-level segmentation of the complete WSI.

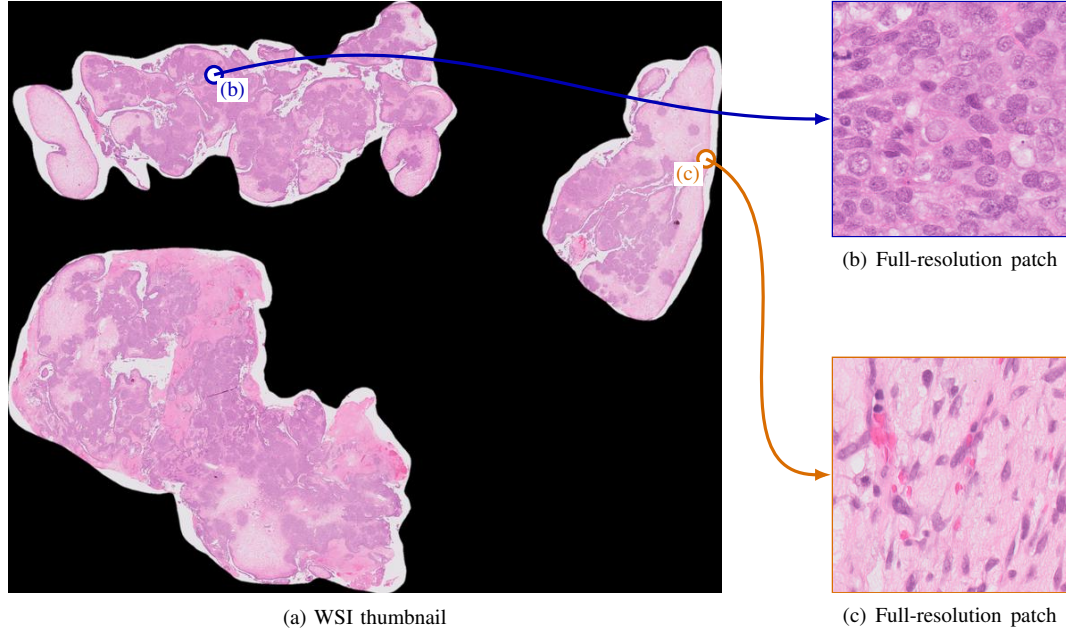


Fig. 1. From a whole-slide image to local model inputs. Panels (b) and (c) show two 256×256 crops from the same WSI in (a). The arrows start at their source regions, located by matching reduced versions of the actual patches against the official WSI thumbnail. The thumbnail illustrates the scale difference; VAE inference used the full-resolution crops.

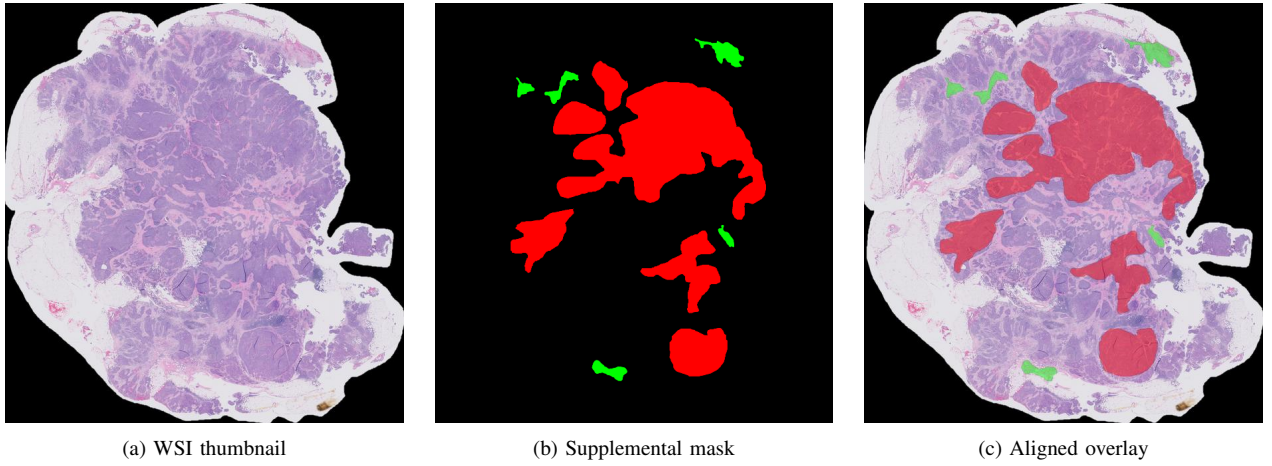


Fig. 2. Actual paired WSI thumbnail and supplemental tissue mask from the UBC-OCEAN training data [15], [16]. The full-resolution WSI and mask measure 40063×42207 pixels; all panels were reduced to 600×632 for display. Red marks tumor, green marks stroma, and blue marks necrosis when present; black in the mask is unpainted and supplies no tissue label. The overlay shows the alignment and partial coverage of the annotations.

D. Data Populations and Partitions

The 513 eligible WSIs were separated at WSI level into two non-overlapping populations (Figure 4). VAE development used 322 training and 39 validation WSIs, contributing 300,000 and 30,000 unlabeled patches. Patch sampling supplied 60,000 training and 6,000 validation patches from each WSI diagnosis; diagnosis was not a VAE training target. WSI counts were not balanced by diagnosis (Table I). The downstream cohort contained 106 training, 23 validation and 23 test WSIs and was fixed before VAE or classifier outputs on that cohort were inspected. Its assignment was shared by both downstream tasks and used diagnosis and annotated-

tissue availability, but not image pixels, VAE representations or model predictions.

For WSI diagnosis, each slide was one labeled MIL bag containing all its historical Otsu-selected foreground patches. Each complete WSI contributed one diagnosis label shared by all patches in its bag. The patch-level tissue classifier instead used high-purity annotated tumor, stroma and necrosis patches. Its training subsets were balanced at the five nested label budgets introduced above, whereas validation and test used fixed, unbalanced task manifests. Table I separates the inputs and class counts of these three stages.

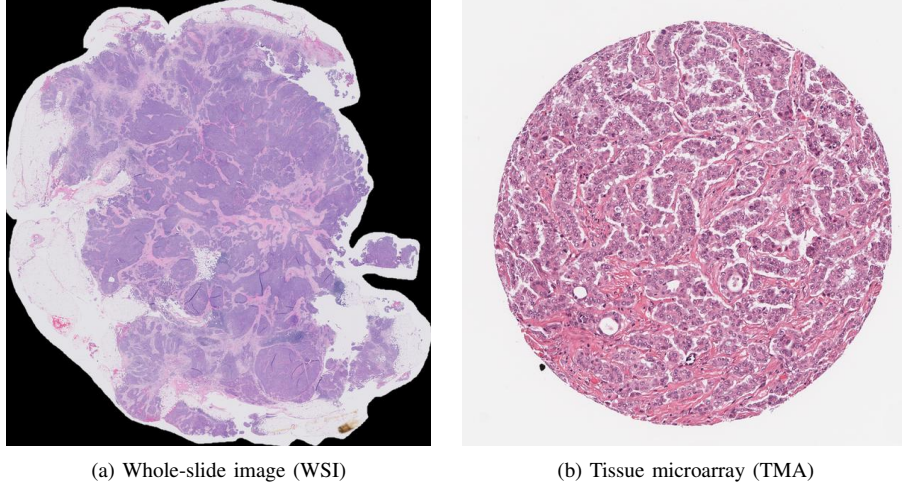


Fig. 3. Actual UBC-OCEAN training images illustrating the two image types [15]. The WSI in (a) measures 40063×42207 pixels at approximately $20\times$ magnification; the TMA in (b) measures 2964×2964 pixels at approximately $40\times$. Both are resized to a similar display height, so the panels do not share a physical scale. The WSI contains about 192 times as many full-resolution pixels and approximately 25,584 versus 121 non-overlapping 256×256 grid cells before foreground filtering. The experiment uses WSIs and excludes TMAs.

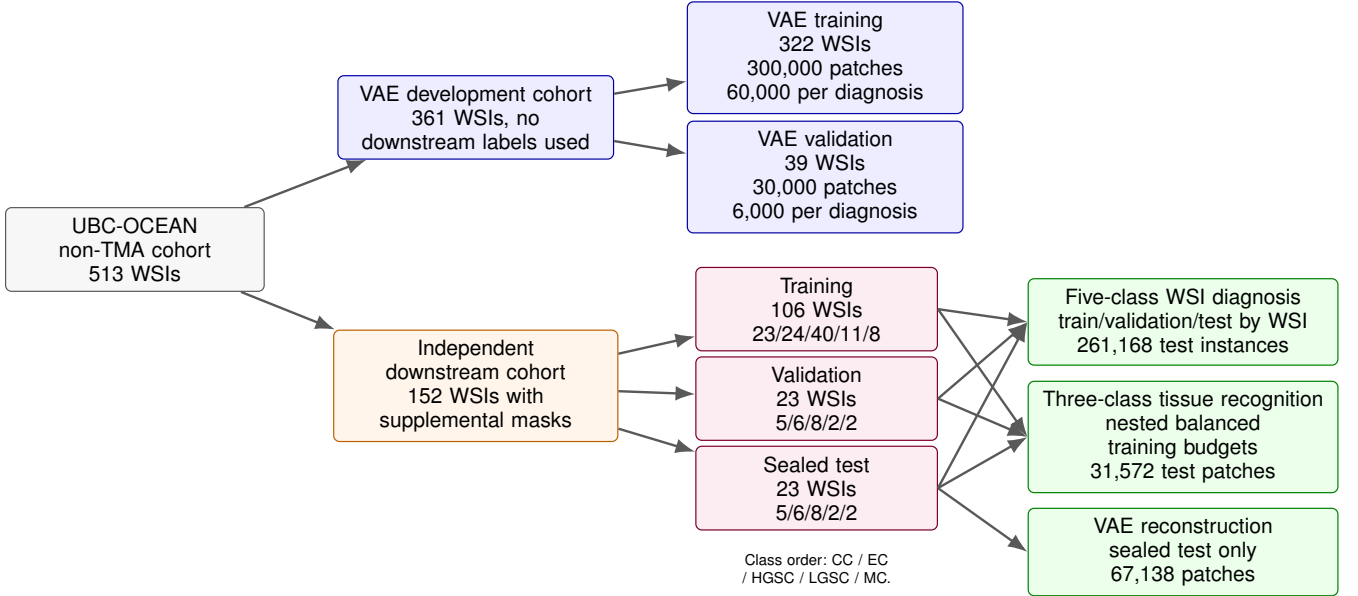


Fig. 4. Data populations and WSI-level partitions. The VAE-development and downstream cohorts are disjoint. All three supervised subsets are fixed at WSI level and shared by the conventional and equivariant representations. Numbers inside split boxes follow the order CC/EC/HGSC/LGSC/MC.

E. Variational Autoencoders and Shared Architecture

An autoencoder learns to turn an image into a smaller representation and then reconstruct the image from it. Its *encoder* extracts and compresses visual features, and its *decoder* uses the compressed embedding to reproduce the input. In a VAE, the encoder specifies a distribution over embeddings rather than one fixed code: here a diagonal Gaussian with mean μ and log-variance $\log \sigma^2$. During training, a latent sample $\mathbf{z}$ is drawn by reparameterization [8]. The distribution is regularized toward a unit Gaussian while the decoder learns image reconstruction. After training, the posterior mean is a deterministic feature map for the classifiers. For reconstruction evaluation, the decoder reconstructs the clean patch from that

mean. Figure 5 shows these paths.

Both models used a four-stage residual encoder informed by ResNet-18 [18]: a stem and two residual blocks at each stage, with $2\times$ fixed anti-aliased downsampling between stages. Their decoder used two residual blocks per stage and reversed the three resolution changes with bilinear upsampling followed by convolution. The spatial sequence was $256 \rightarrow 128 \rightarrow 64 \rightarrow 32$ pixels per side in the encoder and reversed in the decoder. Separate 5×5 posterior heads produced

$$\mu, \log \sigma^2 \in \mathbb{R}^{16 \times 32 \times 32}. \quad (1)$$

The decoder returned a raw three-channel image from a zero-initialized output convolution, without a final tanh. The 16-

TABLE I

INPUTS BY TASK AND SPLIT. DIAGNOSIS ORDER IS CC/EC/HGSC/LGSC/MC; TISSUE ORDER IS TUMOR/STROMA/NECROSIS. VAE PATCH DIAGNOSIS IS INHERITED FROM THE SOURCE WSI, NOT A LOCAL TISSUE LABEL. MIL CLASS COUNTS REFER TO WSIS; TISSUE CLASS COUNTS REFER TO PATCHES.

Task	Split	WSIs	Patches	Class composition
VAE	Train	322	300,000	54/74/144/24/26 WSIs; 60,000 patches per diagnosis.
VAE	Validation	39	30,000	7/9/17/3/3 WSIs; 6,000 patches per diagnosis.
WSI MIL	Train	106	1,224,875	23/24/40/11/8 WSI labels; complete foreground bags.
WSI MIL	Validation	23	264,178	5/6/8/2/2 WSI labels; complete foreground bags.
WSI MIL	Test	23	261,168	5/6/8/2/2 WSI labels; complete foreground bags.
Tissue	Train	106	750–17,013	250–5,671 selected patches per tissue class.
Tissue	Validation	23	31,339	22,255/7,809/1,275 patches.
Tissue	Test	23	31,572	21,796/8,500/1,276 patches.
Reconstruction	Test	23	67,138	Foreground patches; uncertainty aggregated by WSI.

channel spatial embedding preserves coarse locations within the patch; its 32×32 grid is smaller than the original 256×256 input. Table II compares the two implementations.

F. Continuous Rotation Equivariance

A *group* is a collection of transformations that can be composed and inverted. The planar rotation group $SO(2)$ contains every angle from 0 to 360 degrees, with angle addition as its composition rule. A map f is *equivariant* when transforming its input also transforms its output in a predictable way:

$$f(T_g \mathbf{x}) = \rho(g)f(\mathbf{x}), \quad g \in SO(2). \quad (2)$$

Here T_g rotates an image and $\rho(g)$ specifies how the output feature field must rotate. This is distinct from *invariance*, in which the output stays identical after rotation. For a scalar image feature, spatial locations rotate but the value at a corresponding location does not; for a vector feature, its two components rotate as well.

The conventional VAE used ordinary convolutions, GroupNorm and learned scalar gates. Its hidden channel counts were 32, 48, 64 and 96. The experimental $SO(2)$ VAE replaced the convolutional layers with steerable convolutions [5], [6]. A steerable kernel is a filter whose rotated version follows a prescribed rule: rather than learning an unrelated filter for every orientation, the network learns coefficients of radial and angular basis functions that obey the input/output field transformation laws. This construction constrains the convolutional operators; it does not by itself establish that the complete trained encoder obeys Equation 2 exactly under sampled image rotations.

Its hidden features contained F_0 scalar fields and F_1 vector fields. An F_0 copy stores one value per pixel and changes only its spatial location under a rotation. An F_1 copy stores an ordered pair of values per pixel; that pair rotates by the same angle as the image coordinate system. Thus, a layout with n_0 scalar copies and n_1 vector copies has $n_0 + 2n_1$ numerical channels, not $n_0 + n_1$. The four stages used $(n_0, n_1) = (16, 16), (24, 24), (32, 32), (48, 48)$. Field-aware normalization and gates based on the F_1 vector magnitude were used so these operations respect the field types. The precise nonlinearities and normalizations are given below. The posterior heads returned 16 F_0 fields and no F_1

fields, matching the conventional $16 \times 32 \times 32$ posterior. The learned kernel parameters were coefficients of a steerable basis evaluated on fixed pixel grids. The model therefore had fewer learned parameters but more estimated convolutional work (Table II). Figure 6 shows the two residual building blocks before Figure 7 assembles them into the full encoder and decoder schedules.

G. Field Gates and Normalization

The conventional VAE applies a learned scalar gate to each hidden channel, including after residual addition. For a channel value u_c ,

$$A_0(u_c) = u_c \sigma(a_c u_c + b_c), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (3)$$

where a_c and b_c are learned per-channel parameters. The F_0 fields in the $SO(2)$ VAE use the same rule with one parameter pair per scalar field. For an F_1 field $\mathbf{v}_j = (v_{j,1}, v_{j,2})$, the equivariant VAE gates both components by the same factor computed from their rotation-invariant magnitude:

$$r_j = \sqrt{v_{j,1}^2 + v_{j,2}^2 + 10^{-4}}, \quad (4)$$

$$A_1(\mathbf{v}_j) = \mathbf{v}_j \sigma(\alpha_j r_j + \beta_j).$$

The learned α_j, β_j are shared by the two components. A planar rotation changes the vector direction but not r_j , so this gate rotates its output together with the input vector.

Before these gates, the conventional network uses affine GroupNorm with eight groups. For each sample and channel c , it computes the mean and variance over the channels in c 's group and all spatial positions:

$$N_{GN}(u_c) = \gamma_c \frac{u_c - \mu_{G(c)}}{\sqrt{s_{G(c)}^2 + 10^{-5}}} + \delta_c. \quad (5)$$

The equivariant network applies this same eight-group formula only to its F_0 scalar fields. Its F_1 vectors instead use four groups and a shared root-mean-square denominator for both components:

$$N_1(\mathbf{v}_j) = \gamma_j \frac{\mathbf{v}_j}{\sqrt{\text{mean}_{k \in G(j), d \in \{1,2\}, p} v_{k,d,p}^2 + 10^{-5}}}. \quad (6)$$

Here p indexes spatial positions within one sample. The vector rule has no mean subtraction or additive bias, because

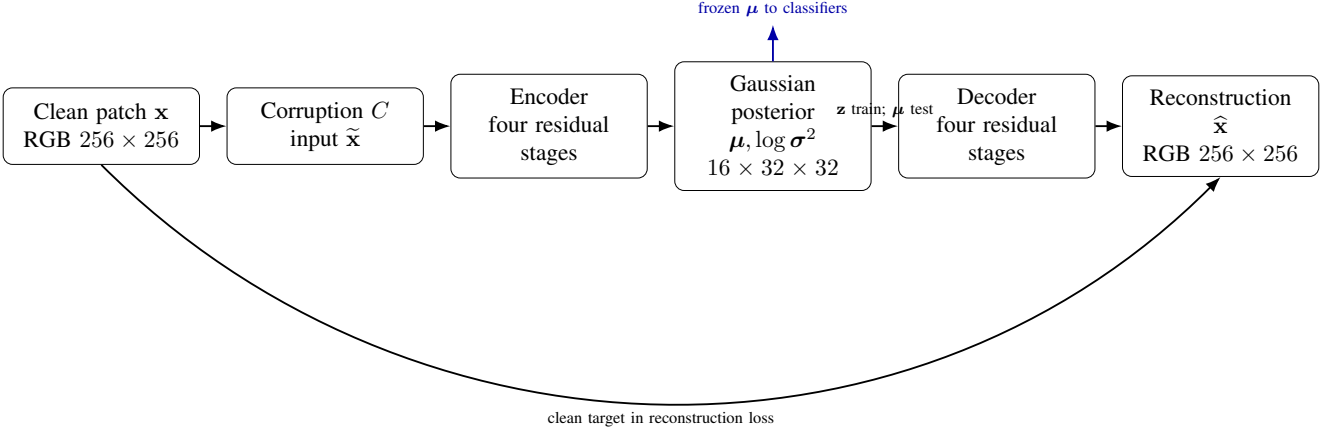


Fig. 5. Shared denoising VAE workflow. Both models have the same input, posterior shape and encoder–decoder resolution schedule. They differ in the type of convolution and the accompanying field normalization and gates. Training reconstructs the clean patch from a possibly corrupted input; classification uses only the frozen posterior mean.

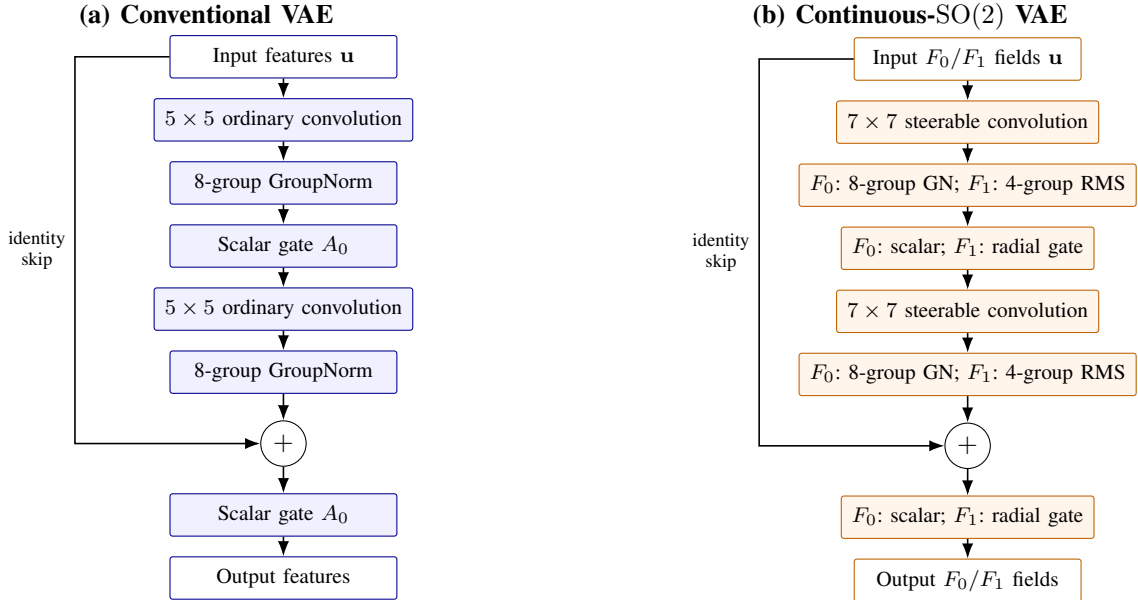


Fig. 6. Same-resolution residual blocks used by the conventional and continuous-SO(2) VAEs. Each main branch contains two learned convolutions; normalization and a gate follow the first convolution, while the second is followed by normalization. The identity skip is added before the output gate. At encoder stage transitions, both branches are downsampled separately with a fixed 5×5 binomial filter: on the main branch after the first gate and on the skip branch before its learned projection and normalization. At decoder transitions, both branches are bilinearly upsampled before their first learned convolution; the skip branch again uses a projection and normalization. The F_0/F_1 field types and gate equations are defined in the surrounding text.

either would introduce a preferred direction. Its denominator is unchanged by rotations in feature space. The normalization statistics and gates are evaluated in FP32 in the SO(2) implementation.

H. VAE Objective and Training

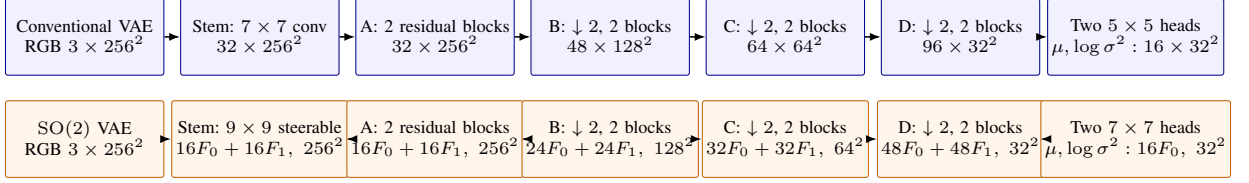
Both models used a denoising VAE training criterion: an encoder sees a possibly corrupted patch, and the decoder is trained to reconstruct the corresponding clean patch [9]. In our implementation, with probability 0.3 the corruption operator C perturbed hematoxylin/eosin/residual stain coordinates and

added image-space Gaussian noise [19]. The resulting input was clamped to the normalized $[-1, 1]$ RGB domain. A patch could therefore be clean or corrupted at the input, while its reconstruction target was always the same clean patch:

$$\mathbf{x} \xrightarrow{C} \tilde{\mathbf{x}} \xrightarrow{q_\phi} \mathbf{z} \xrightarrow{D_\theta} \hat{\mathbf{x}} \quad \text{compared with } \mathbf{x}. \quad (7)$$

The encoder distribution $q_\phi(\mathbf{z} | \tilde{\mathbf{x}})$ is the diagonal Gaussian determined by the mean and log-variance heads; D_θ is the decoder. During training, $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

(a) Encoder paths



(b) Decoder paths

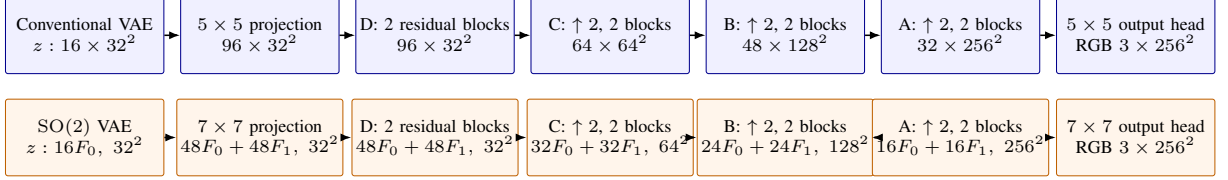


Fig. 7. Layer-level architecture of the two frozen VAEs. The residual building blocks are detailed in Figure 6; stage-changing blocks additionally resample both branches and project the skip branch. A conventional hidden width is a channel count; in the SO(2) model, $nF_0 + nF_1$ means n scalar fields and n two-component vector fields. Posterior mean and log-variance use separate heads; training samples $\mathbf{z}$, whereas downstream classifiers use μ . The conventional stem uses a 7×7 kernel and its other convolutions use 5×5 kernels; the SO(2) stem uses a 9×9 steerable kernel and its other convolutions use 7×7 kernels.

TABLE II
ARCHITECTURE COMPARISON.

Property	Conventional VAE	Continuous-SO(2) VAE
Hidden representation	Ordinary channels: 32, 48, 64, 96	(F_0, F_1) copies: (16,16), (24,24), (32,32), (48,48); 48, 72, 96, 144 numerical channels
Learned convolutions	43 ordinary convolutions	43 steerable convolutions
Normalization/nonlinearity	Eight-group GroupNorm; learned scalar gate	Eight-group F_0 GroupNorm; four-group F_1 RMS normalization; scalar and radial gates
Posterior	$\mu, \log \sigma^2 \in \mathbb{R}^{16 \times 32 \times 32}$	Same spatial Gaussian posterior, represented by scalar fields
Learned parameters	3,958,435	1,180,035
Estimated convolutional work	3.65×10^{10} MAC/sample	1.598×10^{11} MAC/sample
Rotation constraint	None beyond translation-equivariant convolution	Continuous SO(2) kernel constraint

The implemented loss was

$$\mathcal{L} = \text{MAE}(\hat{\mathbf{x}}, \mathbf{x}) + 0.1[1 - \text{SSIM}(\hat{\mathbf{x}}, \mathbf{x})] + \beta D_{\text{KL}}(q_{\phi}(\mathbf{z} | \hat{\mathbf{x}}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})), \quad (8)$$

where MAE measures pixelwise reconstruction error, SSIM measures structural similarity [20], and the KL term regularizes the input-dependent posterior toward a unit Gaussian. The KL weight increased linearly from zero to 0.01 during the first 6,000 updates and remained fixed thereafter.

AdamW [21] used coefficients (0.9, 0.999), $\epsilon = 10^{-8}$, weight decay 10^{-5} and gradient-norm clipping at 1.0. The learning rate increased linearly for 600 updates to 10^{-3} and then decreased with a cosine schedule to 10^{-5} . Each model was trained for 60,000 successful updates with a global batch of 50 patches distributed over two NVIDIA T4 GPUs. Validation and checkpointing occurred every 3,000 updates. The comparison used the final update-60,000 checkpoint for both models rather than selecting different VAE checkpoints from the downstream results.

After VAE training, all encoder and decoder parameters were frozen. Downstream tasks used the posterior mean μ

rather than a sampled latent. For every selected coordinate, both encoders received the same RGB patch. The resulting paired latent datasets had identical WSI membership, tissue labels and patch coordinates.

I. Reconstruction Evaluation

The reconstruction test contained 67,138 foreground patches from the 23 test WSIs. For each slide, eligible foreground coordinates were sorted by row and column (y , then x). If there were more than 3,000, 3,000 evenly spaced rank positions were selected across that ordered list; otherwise all were retained. Thus, 22 slides contributed 3,000 patches each and one contributed its available 1,138. The cap limited the influence of slides with many more eligible patches. MAE and mean squared error (MSE) were measured in the normalized $[-1, 1]$ domain. Peak signal-to-noise ratio (PSNR) and SSIM were measured after conversion to $[0, 1]$ and clamping. Mean MAE was the primary reconstruction endpoint. The other metrics describe complementary error properties.

J. Classification A: Five-Class WSI Diagnosis

The diagnostic task predicted the five ovarian-cancer subtypes. Every selected foreground patch was represented by its frozen posterior mean. A trainable convolutional patch encoder compressed the $16 \times 32 \times 32$ latent map. Two local attention blocks exchanged information within a radius-two spatial neighborhood, and a global stream summarized the complete WSI through learned sigmoid attention. A final classification layer produced five logits. The supervised model contained approximately 1.51 million parameters.

Training used one complete WSI bag per optimizer step, AdamW and class-weighted cross-entropy computed from the 106 training WSIs. Each representation branch started from the same classifier weights and received the same WSI order. The maximum was 150 epochs, validation was performed twice per epoch, and each branch independently retained its highest validation macro-F1 checkpoint. The primary test metric was macro-F1; balanced accuracy, accuracy, mean cross-entropy, classwise F1 and confusion matrices were secondary summaries.

K. Classification B: Tissue Recognition and Label Budgets

This complementary classification setting removes WSI aggregation and uses the complete spatial embedding of one patch. It tests how readily a compact supervised model can recover local tissue information from each frozen representation as the label budget changes. The task predicted tumor, stroma or necrosis from a single frozen latent map. The classifier contained three convolutional stages $16 \rightarrow 32 \rightarrow 64 \rightarrow 128$, GroupNorm, GELU activations [22], adaptive average pooling and a linear $128 \rightarrow 3$ output layer.

Training subsets contained 250, 500, 1,000, 2,500 and 5,671 labeled patches per class. They were class balanced, nested and selected without replacement; every smaller set was a subset of each larger one. WSI-stratified ordering ensured that the smallest subset did not collapse onto a single slide. Each normal/equivariant pair started from identical classifier weights and received the same patch order. Models were optimized for at most 30 epochs, and the best validation macro-F1 checkpoint was retained. The test distribution was not balanced and contained 21,796 tumor, 8,500 stroma and 1,276 necrosis patches from 23 WSIs.

L. Statistical Analysis

All difference intervals are reported as conventional-minus-equivariant. Reconstruction differences used 10,000 paired cluster-bootstrap replicates at WSI level [23]. WSI diagnosis used 10,000 paired bootstrap replicates stratified by diagnosis. Tissue recognition used a paired WSI bootstrap stratified by observed tissue support. The five label-budget macro-F1 differences and the area under the log-budget learning curve (AULC) difference were considered together as six related comparisons. Their 95% intervals were adjusted simultaneously: across the six comparisons, the bootstrap procedure targets 95% coverage for the whole set of differences. A difference interval entirely below zero favors the equivariant

TABLE III
RECONSTRUCTION ON 67,138 TEST PATCHES. VALUES ARE MEAN (POPULATION STANDARD DEVIATION).

Metric	Conventional	SO(2)
MAE ↓	0.0629 (0.0239)	0.0642 (0.0223)
MSE ↓	0.0080 (0.0074)	0.0083 (0.0067)
PSNR dB ↑	27.923 (2.721)	27.658 (2.595)
SSIM ↑	0.7575 (0.0842)	0.7436 (0.0862)

TABLE IV
OVERALL WSI DIAGNOSIS RESULTS ON 23 TEST SLIDES. DIFFERENCE INTERVALS ARE CONVENTIONAL-MINUS-SO(2).

Metric	Conventional	SO(2)	Difference 95% CI
Macro-F1 ↑	0.389	0.483	$[-0.207, 0.050]$
Balanced accuracy ↑	0.510	0.503	$[-0.167, 0.180]$
Accuracy ↑	0.522	0.609	$[-0.217, 0.043]$
Mean cross-entropy ↓	1.323	0.993	$[0.135, 0.524]$

model when higher is better; an interval entirely above zero favors the conventional model. An interval crossing zero leaves that comparison unresolved. Patch standard deviations describe variation among patches, while confidence intervals resample WSIs as the biological units.

IV. RESULTS

A. VAE Reconstruction

Figure 8 displays clean inputs and both models' reconstructions for 25 fixed validation patches.

The conventional VAE had lower pooled MAE and MSE and higher pooled PSNR and SSIM (Table III). Mean MAE was 0.062872 for the conventional model and 0.064230 for the equivariant model. The paired difference was -0.001358 , with a 95% interval of $[-0.002213, 0.000046]$. The interval included zero, so the primary analysis did not establish a reconstruction difference. The remaining metrics are descriptive secondary summaries.

B. Classification A: Five-Class WSI Diagnosis

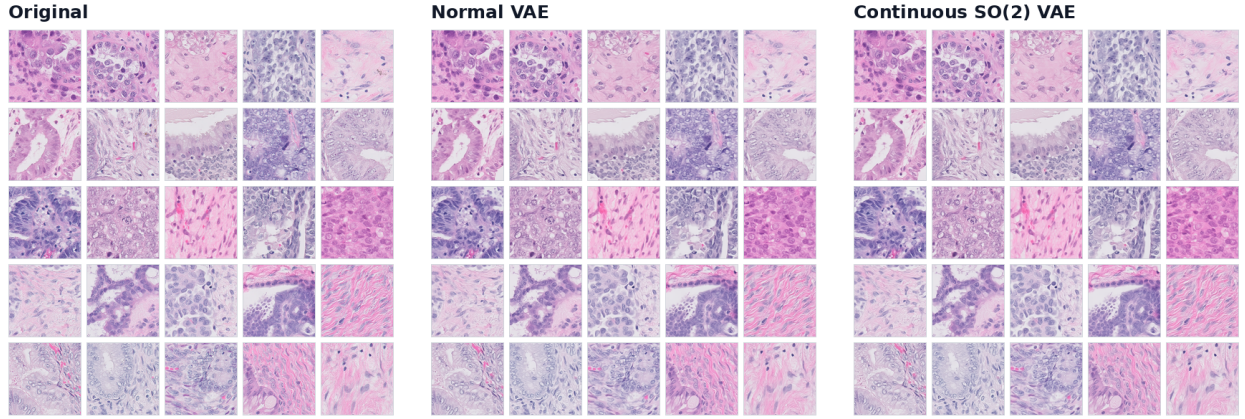
Macro-F1 was 0.483 for the equivariant representation and 0.389 for the conventional representation (Table IV). The conventional-minus-equivariant difference was -0.095 , with a 95% interval of $[-0.207, 0.050]$. The interval included zero. Among the secondary metrics, the equivariant representation had lower mean cross-entropy, with a WSI-resampling interval excluding zero. The macro-F1 interval still crossed zero.

Classwise point estimates varied (Table V). The conventional representation had zero F1 for EC and LGSC and correctly classified both MC slides; the equivariant representation had higher EC and LGSC F1 and no correct MC prediction. With only two test WSIs in each of LGSC and MC, these contrasts are unstable.

C. Classification B: Three-Class Tissue Recognition

The tissue results changed with the amount of labeled training data (Figure 9 and Table VI). The equivariant representation had higher macro-F1 at 250, 500, 1,000 and

Final reconstructions — same predetermined fixed validation 25



Ordered by frozen selector rank (five patches per source histotype); FP16 archives shown after $[-1,1] \rightarrow [0,1]$ clamping.

Fig. 8. Original patches and final reconstructions for the same 25 fixed VAE validation examples. The quantitative reconstruction results use 67,138 patches from the 23 test WSIs.

TABLE V
CLASSWISE WSI F1 AND TEST SUPPORT.

Diagnosis	Support	Conventional F1 $\uparrow$	SO(2) F1 $\uparrow$
CC	5	0.667	0.750
EC	6	0.000	0.533
HGSC	8	0.706	0.800
LGSC	2	0.000	0.333
MC	2	0.571	0.000

2,500 labels per class. At 500 labels per class, the macro-F1 values were 0.575 and 0.511, and the simultaneous interval for conventional-minus-equivariant was $[-0.121, -0.008]$. This was the only label budget whose simultaneous interval excluded zero.

At 500 labels per class, the classwise F1 values for conventional/equivariant representations were 0.769/0.781 for tumor, 0.485/0.560 for stroma and 0.279/0.384 for necrosis. These values show how the aggregate macro-F1 difference was distributed across the three tissue classes.

At the largest budget, the conventional representation obtained macro-F1 0.766 compared with 0.710 for the equivariant representation. The simultaneous interval included zero, with a lower bound of -0.001 . The corresponding classwise F1 values were 0.868/0.871 for tumor, 0.728/0.711 for stroma and 0.701/0.548 for necrosis. The normalized area under the log-budget learning curve was 0.615 for the conventional representation and 0.631 for the equivariant representation; its simultaneous difference interval included zero.

V. DISCUSSION

The ordering of the two representations depended on the endpoint and tissue label budget. The conventional VAE had lower pooled MAE/MSE and higher pooled PSNR/SSIM point estimates, whereas the equivariant representation had the

TABLE VI
TISSUE MACRO-F1 ON 31,572 TEST PATCHES. SIMULTANEOUS INTERVALS ARE REPORTED FOR CONVENTIONAL-MINUS-SO(2).

Labels/class	Conventional $\uparrow$	SO(2) $\uparrow$	Simultaneous 95% CI
250	0.508	0.523	$[-0.072, 0.041]$
500	0.511	0.575	$[-0.121, -0.008]$
1,000	0.582	0.598	$[-0.072, 0.041]$
2,500	0.701	0.714	$[-0.069, 0.044]$
5,671	0.766	0.710	$[-0.001, 0.112]$

higher WSI macro-F1 point estimate and higher tissue macro-F1 at the four smaller budgets. Most primary or familywise intervals included zero. The single tissue-budget interval that excluded zero applies to that contrast only and does not establish a monotone or general label-efficiency effect. The reconstruction point-estimate ordering was not consistent with the downstream point-estimate ordering.

Patch-based WSI systems depend on both the instance representation and the bag aggregator. Prior MIL work has developed the representation, aggregation or both [10]–[13]. This experiment held the downstream protocol fixed while changing the frozen representation. The WSI interval remained wide, and classwise estimates were unstable for diagnoses with two test slides. The results characterize the two implemented representation branches but do not establish a general WSI or tissue-classification advantage.

The models also differ in capacity and computation. The equivariant VAE has approximately 30% of the learned parameters of the conventional VAE, but its kernel expansion requires more estimated multiply-accumulate operations. Any difference may reflect equivariance, parameter count, optimization or their interaction. The experiment does not separate these mechanisms.

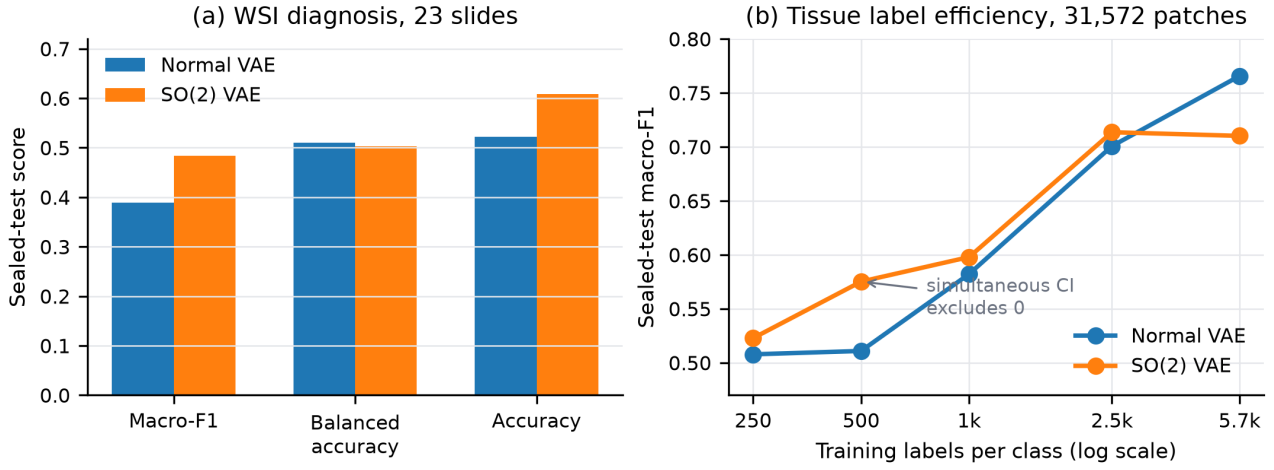


Fig. 9. Frozen-representation classification results. (a) Five-class WSI diagnosis on 23 test slides. (b) Tissue macro-F1 on 31,572 test patches under nested class-balanced training budgets. At 500 labels per class, the adjusted 95% interval for the conventional-minus-equivariant macro-F1 difference is entirely negative.

VI. LIMITATIONS

Each VAE and downstream configuration was represented by one training trajectory. The bootstrap intervals quantify variation across the observed WSIs, not variation across new initializations, data orders or optimization runs. The WSI test contains 23 slides, including two each for LGSC and MC. Tissue recognition contains many patches but only 23 biological clusters, and necrosis is present in five of them.

Supplemental tissue masks are incomplete. The tissue experiment evaluates selected high-purity annotated regions and cannot be interpreted as exhaustive segmentation of complete WSIs. The study uses one ovarian-cancer data source.

VII. CONCLUSION

We compared conventional and continuous-SO(2) VAEs as frozen histopathology representations under a common experimental protocol. With one training trajectory per configuration and intervals limited to WSI resampling, the estimates varied by endpoint and label budget. The experiment did not establish a general advantage for either implemented representation; repeated seeds and capacity-matched controls are needed to isolate the contribution of continuous rotation equivariance.

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